Supplementary material

Separating reconstruction, generation and antibacterial activity in antimicrobial peptide design

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Supplementary Methods describe the model architecture, evaluation metrics and antibacterial assays. Tables S1–S5 provide computational results and peptide-level measurements; Table S6 describes the published AMPGen workflow. Original MIC summaries and retrospective algorithmic endpoints are presented separately. Full-precision computational and assay data accompany the tables.

Supplementary Methods

Network architecture

The encoder embeds 21 input symbols (20 residues and padding) into 100 channels over 48 positions. Three residue-aligned descriptors are projected from 3 to 100 channels by a width-3 convolution and condition an eight-layer gated, weight-normalized convolutional encoder (100 channels, kernel width 5, dilation 1). A nine-dimensional global descriptor is fused by scaled dot-product attention, followed by a width-1 projection to 32 channels, split into 16-channel posterior means and log standard deviations. Each posterior parameter therefore has shape 16×48. The nonsqueezed decoder has four gated convolutional layers with 16 channels, kernel width 5 and dilation 1, followed by a width-1 projection to 21 raw scores per position (or one scalar score for the scalar comparator). This path uses no positional embedding or temporal downsampling.

The prior transform comprises four mean-only additive residual-coupling layers alternating with four channel flips. Each coupling splits the 16 latent channels into two groups of eight and uses a 64-channel, four-layer gated convolutional conditioner (kernel width 3, dilation 1), conditioned on the three residue descriptors. The transform contributes to the KL density term; reconstruction and generation decode posterior samples directly, without inversion of the prior transform.

The diffusion denoiser receives the stacked posterior mean and log standard deviation as a two-channel 16×48 array. Its two-dimensional U-Net has three downsampling stages (channel widths $64 \to 128 \to 256 \to 256$), a $256 \to 512 \to 256$ bottleneck, three upsampling stages with skip connections (128, 64 and 64 output channels), and a final 1×1 convolution to two channels. Four-head self-attention follows each downsampling and upsampling stage. A 256-dimensional sinusoidal timestep embedding is added to a 256-dimensional embedding of the two terminal states and projected into each downsampling and upsampling stage. The additional hierarchical embedding tables defined in the inherited class are not used in this forward path. Training drops the terminal label with probability 0.1; the recorded generation path uses conditional predictions without classifier-free guidance interpolation.

These architecture details follow the executed configuration and active forward path. `source/architecture_facts.json` binds each component to packaged source files, line ranges and SHA-256 hashes. Registered parameter totals include parameters unused in this forward path.

Molecular descriptors

The per-residue descriptor triplet is ordered as RDKit MolWt, MolLogP and TPSA. The source constructs each canonical L residue as an isolated free amino acid, rounds these three outputs to two decimal places and uses the same lookup for both peptide terminal states. The nine global descriptors, in order, are RingCount, NumAromaticRings, NumSaturatedRings, FractionCSP3, HallKierAlpha, LabuteASA, MaxPartialCharge, MinPartialCharge and TPSA. They are the native outputs of the correspondingly named RDKit descriptor functions on the full peptide, with no manual unit conversion or rounding in this source.

The molecular construction uses `Chem.MolFromSequence` with `flavor=0`. FREE retains the free C terminus; AMD changes the terminal OXT oxygen to nitrogen and validates the resulting amide, element counts, exact-mass change and residue chirality. This construction models database-declared terminal chemistry. The accompanying `data/train_dev/descriptor_dictionary.json` specifies the field order and binds the retained source by SHA-256. The formal reconstruction and warmup scoring contracts identify this source. The numerical input CSVs remain unchanged; this revision documents the source definitions without independently regenerating each descriptor value. Training-derived standardization is recorded separately in the supplied scaler files.

Execution configuration

The VAE optimizer was Adam with base learning rate 5×10^{-4}, a 1,000-step warmup, floor 10^{-7}, coefficients $(0.9, 0.999)$, epsilon 10^{-8} and zero weight decay. We used the step-offset convention specified in the accompanying configuration. Gradient norm was clipped at one. Diffusion used AdamW with maximum learning rate 0.005, epsilon 10^{-5}, coefficients $(0.9, 0.999)$, weight decay 0.01 and a 1,000-step OneCycleLR schedule. The EMA coefficient was 0.995, with the implementation copying parameters during these first 1,000 updates. The 999-element linear beta schedule ranged from 10^{-4} to 0.02; training and reverse timesteps ran from 1 through 998. The complete configuration is supplied with the model implementation.

The VAE and diffusion phase ceilings were 1,920 and 1,020 seconds within the common 2,940-second training ceiling. We monitored training every 100 updates and used the final scheduled checkpoint. Each arm used one training seed; the development partition was also used during model development. Configuration and dataset hashes accompany the implementation.

Comparator monitoring

We trained the LSTM for 1,000 updates and used the final checkpoint. Its development negative log-likelihood was 2.6153 at update 200 and 3.1698 at the final update. The common update counts and time ceilings define the computational protocol; model capacities and realized computation differ between the LSTM and diffusion pipelines.

Distribution statistics

For a sequence `x`, let `f(x)` concatenate length/48, the 20 residue fractions and the 400 dipeptide fractions, each block multiplied by $1/\sqrt{3}$. The reported biased linear-kernel statistic is

$$\mathrm{MMD}^2 = \left\| \frac{1}{n} \sum_i f(x_i) - \frac{1}{m} \sum_j f(r_j) \right\|_2^2$$

For a one-residue sequence the dipeptide block is zero. This measures a fixed collection of low-order features, not complete distributional agreement. Additional distances are residue and dipeptide Jensen–Shannon divergence in bits, their respective L_1 distances, and the length Kolmogorov–Smirnov

statistic. Residue and dipeptide JS, L_1 and entropy use pooled counts, unlike the per-sequence normalization followed by averaging used for MMD. References use the corresponding terminal group in training or development, and entropy discrepancies and exact training-copy counts are retained as complementary diagnostics.

Retrospective assay computation

For each sample, strain and replicate, the endpoint is the lowest tested concentration with $OD_{600} < 0.10$. The ten two-fold concentrations are 1.953125, 3.90625, 7.8125, 15.625, 31.25, 62.5, 125, 250, 500 and 1,000 μg/mL. A series with no qualifying concentration has no numeric endpoint. The algorithm takes a numeric value when at least two of three replicate calls agree. It retains `no_endpoint_majority` when at least two replicates have no endpoint and `no_consensus` when the three calls have no majority. For comparison with the original summary, both nonnumeric states are mapped to its N code. The retrospective rule is analyzed alongside the original MIC values rather than replacing them.

The original summary contains 30 numeric and 78 N cells. The algorithm produces 30 numeric-consensus cells, 77 no-endpoint-majority cells and one no-consensus cell. Table S5b presents all 108 paired endpoints, and Table S5c gives the three differing peptide–strain combinations with their replicate calls. Activity fractions use all 27 assayed peptides as the denominator. N denotes a nonnumeric outcome and is not interpreted as confirmed absence of inhibition. Table S5f retains the complete threshold analysis.

S1 Reconstruction records (Table S1)

Reconstruction uses the posterior mean. Residue micro accuracy divides aligned correct target positions by all nonpadding target residues. Per-amino-acid recall divides aligned correct positions for that residue by its target support. Whole-sequence equality and length error remain separate measures. The three models use the same 279 training and 41 development targets, yielding 960 saved reconstruction records.

Table S1a. Complete reconstruction summary. Micro accuracy excludes padding; AA macro is the unweighted mean recall of supported amino-acid classes.

Model	Split	Peptides	Residues	Correct	Micro	AA macro	Length MAE
CE21	train	279	4868	3580	0.7354	0.4646	0.0681
MSE21	train	279	4868	585	0.1202	0.0393	4.5806
Scalar	train	279	4868	580	0.1191	0.0752	1.3262
CE21	development	41	701	497	0.7090	0.4276	0.0000
MSE21	development	41	701	88	0.1255	0.0357	5.5610
Scalar	development	41	701	111	0.1583	0.0805	1.3415

Table S1c (train). Complete per-amino-acid support, correct counts and recall.

AA	Support	CE correct	MSE correct	Scalar correct	CE recall	MSE recall	Scalar recall
A	341	297	0	0	0.8710	0.0000	0.0000
C	43	0	0	0	0.0000	0.0000	0.0000
D	47	0	0	0	0.0000	0.0000	0.0000
E	60	0	0	0	0.0000	0.0000	0.0000

AA	Support	CE correct	MSE correct	Scalar correct	CE recall	MSE recall	Scalar recall
F	274	271	0	79	0.9891	0.0000	0.2883
G	311	262	0	75	0.8424	0.0000	0.2412
H	97	0	0	6	0.0000	0.0000	0.0619
I	395	118	0	2	0.2987	0.0000	0.0051
K	803	795	425	298	0.9900	0.5293	0.3711
L	622	562	158	2	0.9035	0.2540	0.0032
M	47	0	0	2	0.0000	0.0000	0.0426
N	114	0	0	1	0.0000	0.0000	0.0088
P	190	88	0	8	0.4632	0.0000	0.0421
Q	90	0	0	17	0.0000	0.0000	0.1889
R	571	571	2	73	1.0000	0.0035	0.1278
S	167	167	0	5	1.0000	0.0000	0.0299
T	111	30	0	1	0.2703	0.0000	0.0090
V	275	185	0	0	0.6727	0.0000	0.0000
W	236	234	0	7	0.9915	0.0000	0.0297
Y	74	0	0	4	0.0000	0.0000	0.0541

Table S1c (development). Complete per-amino-acid support, correct counts and recall.

AA	Support	CE correct	MSE correct	Scalar correct	CE recall	MSE recall	Scalar recall
A	38	26	0	0	0.6842	0.0000	0.0000
C	3	0	0	0	0.0000	0.0000	0.0000
D	9	0	0	0	0.0000	0.0000	0.0000
E	9	0	0	0	0.0000	0.0000	0.0000
F	47	47	0	15	1.0000	0.0000	0.3191
G	56	42	0	9	0.7500	0.0000	0.1607
H	8	0	0	0	0.0000	0.0000	0.0000
I	54	9	0	0	0.1667	0.0000	0.0000
K	142	142	69	68	1.0000	0.4859	0.4789
L	91	86	17	1	0.9451	0.1868	0.0110
M	3	0	0	0	0.0000	0.0000	0.0000
N	17	0	0	0	0.0000	0.0000	0.0000
P	21	8	0	0	0.3810	0.0000	0.0000
Q	17	0	0	6	0.0000	0.0000	0.3529
R	49	49	2	8	1.0000	0.0408	0.1633
S	28	28	0	2	1.0000	0.0000	0.0714
T	15	2	0	0	0.1333	0.0000	0.0000
V	61	30	0	1	0.4918	0.0000	0.0164
W	28	28	0	1	1.0000	0.0000	0.0357
Y	5	0	0	0	0.0000	0.0000	0.0000

Table S1b is supplied as an editable 960-row CSV containing every saved reconstruction (279 training and 41 development peptides for each of three models), including target/predicted sequences, row ID, terminal condition, aligned correct positions, lengths, per-peptide accuracy and exact-match status.

S2 Generation comparisons (Table S2)

All values below use canonical outputs of length 1–48 and the corresponding terminal reference group. Each generated group contains 16 requests. The six statistics are fixed-feature biased linear MMD squared, residue Jensen–Shannon divergence, residue L_1 distance, dipeptide Jensen–Shannon divergence, dipeptide L_1 distance and the length Kolmogorov–Smirnov statistic. Divergences use base-two logarithms; no significance test is attached. Lower means closer under that statistic, not more biologically active.

The length cap has different semantics: LSTM generation can reach the cap without sampling EOS; CE21 and MSE21 select a prefix length without sampling EOS; scalar decoding uses the first padding category. An execution status of "ok" does not imply natural termination for every model.

Table S2a (train). All six distribution distances; each model–terminal group contains 16 generation requests. Lower is closer under the statistic.

Model	Terminal	MMD^2	AA JS	AA L_1	Pair JS	Pair L_1	Length KS
CE21	Free	0.01742	0.10169	0.51757	0.33968	1.01911	0.27976
CE21	Amide	0.00895	0.09447	0.48963	0.29311	0.93571	0.25531
MSE21	Free	0.07970	0.28258	0.94077	0.54373	1.44107	0.47321
MSE21	Amide	0.09220	0.26188	0.92273	0.52182	1.39526	0.44118
Scalar	Free	0.03521	0.19289	0.82623	0.58372	1.44584	0.39484
Scalar	Amide	0.03290	0.17476	0.81553	0.55988	1.44251	0.59477
terminal_LSTM	Free	0.00409	0.02824	0.25980	0.24461	0.81003	0.13393
terminal_LSTM	Amide	0.00316	0.02143	0.19005	0.24480	0.82108	0.18219

Table S2a (development). All six distribution distances; each model–terminal group contains 16 generation requests. Lower is closer under the statistic.

Model	Terminal	MMD^2	AA JS	AA L_1	Pair JS	Pair L_1	Length KS
CE21	Free	0.01733	0.09932	0.53532	0.41789	1.14371	0.26389
CE21	Amide	0.00771	0.08845	0.42518	0.36559	1.06934	0.30163
MSE21	Free	0.08233	0.29555	1.02041	0.57409	1.47635	0.45833
MSE21	Amide	0.08485	0.25872	0.88895	0.51038	1.36652	0.61957
Scalar	Free	0.03988	0.20632	0.81437	0.68353	1.57098	0.53472
Scalar	Amide	0.03133	0.17014	0.76550	0.65774	1.51300	0.60870
terminal_LSTM	Free	0.01157	0.05963	0.44325	0.37162	1.01636	0.20139
terminal_LSTM	Amide	0.00736	0.04485	0.32838	0.30612	0.91346	0.24457

Table S2b. All 24 aligned method comparisons (two terminal conditions, two reference sets, six statistics).

Terminal	Reference	Statistic	CE21	MSE21	Scalar	LSTM
Free	Train	AA JS	0.10169	0.28258	0.19289	0.02824
Free	Train	AA L_1	0.51757	0.94077	0.82623	0.25980
Free	Train	Pair JS	0.33968	0.54373	0.58372	0.24461
Free	Train	Pair L_1	1.01911	1.44107	1.44584	0.81003
Free	Train	Length KS	0.27976	0.47321	0.39484	0.13393
Free	Train	MMD^2	0.01742	0.07970	0.03521	0.00409

Terminal	Reference	Statistic	CE21	MSE21	Scalar	LSTM
Free	Dev	AA JS	0.09932	0.29555	0.20632	0.05963
Free	Dev	AA L_1	0.53532	1.02041	0.81437	0.44325
Free	Dev	Pair JS	0.41789	0.57409	0.68353	0.37162
Free	Dev	Pair L_1	1.14371	1.47635	1.57098	1.01636
Free	Dev	Length KS	0.26389	0.45833	0.53472	0.20139
Free	Dev	MMD^2	0.01733	0.08233	0.03988	0.01157
Amide	Train	AA JS	0.09447	0.26188	0.17476	0.02143
Amide	Train	AA L_1	0.48963	0.92273	0.81553	0.19005
Amide	Train	Pair JS	0.29311	0.52182	0.55988	0.24480
Amide	Train	Pair L_1	0.93571	1.39526	1.44251	0.82108
Amide	Train	Length KS	0.25531	0.44118	0.59477	0.18219
Amide	Train	MMD^2	0.00895	0.09220	0.03290	0.00316
Amide	Dev	AA JS	0.08845	0.25872	0.17014	0.04485
Amide	Dev	AA L_1	0.42518	0.88895	0.76550	0.32838
Amide	Dev	Pair JS	0.36559	0.51038	0.65774	0.30612
Amide	Dev	Pair L_1	1.06934	1.36652	1.51300	0.91346
Amide	Dev	Length KS	0.30163	0.61957	0.60870	0.24457
Amide	Dev	MMD^2	0.00771	0.08485	0.03133	0.00736

Table S2d. Eligibility, exact training copies and absolute entropy gaps in bits.

Model	Terminal	Requests	1–48	5–48	Copies	AA gap	Pair gap
CE21	Amide	16	16	15	0	0.5628	1.1944
CE21	Free	16	16	14	0	0.6667	1.6514
MSE21	Amide	16	16	10	0	1.7851	3.3978
MSE21	Free	16	16	16	0	1.9200	3.8308
Scalar	Amide	16	16	16	0	0.3635	0.2643
Scalar	Free	16	16	16	0	0.0307	0.8531
terminal_LSTM	Amide	16	16	16	0	0.2423	0.9979
terminal_LSTM	Free	16	16	16	0	0.1575	0.8012

Table S2c retains all 96 reported values, independently recomputed values and absolute differences; the maximum absolute difference is 2.45×10^{-15} (rounded upward). Table S2e contains independently recounted validity, uniqueness and exact-copy totals.

S3 Activity-regression experiments (Table S3)

Data semantics and group construction

We used a separate DBAASP development cache for activity regression. These data were analyzed separately from the generative training partition and were not used to label the main comparison's generated sequences. We analyzed concentrations in μg/mL and μM separately, retaining distinct encodings for point values, explicit bounds, ambiguous ranges and plus/minus formats.

Operational sequence families connected uppercased backbones when their unit-cost Levenshtein distance was at most 20% of the longer sequence length; transitive components define the family. Source groups further joined families sharing a PubMed ID, or a normalized title plus publication year when a PubMed ID was absent. Uppercasing serves only this conservative grouping operation,

not a claim of identical stereochemistry or a label conversion. Operational sequence families and source–family connected components were used to keep related peptide and source records together in validation. Entire groups were left out, with no split of repeated records for the same peptide between training and validation. These components are a leakage-control rule, not verified biological independence. Within a training set, each source group has equal weight, each peptide within a group equal weight, and each record within a peptide equal weight. The primary MAE averages absolute log10 errors in that order. The corresponding RMSE is the square root of the identically weighted mean squared error, not the average of fold RMSEs.

Native point-value baselines

The first experiment used 46 mass-unit points from 33 peptide IDs, 17 operational families and 10 source groups, and 39 molar-unit points from 13 IDs, eight families and seven groups. A training-weighted mean, a composition ridge and a convolutional network were evaluated in 17 group-held-out folds. Ridge uses 20 amino-acid fractions and log length, training-only standardization, an unpenalized intercept and penalty coefficient one.

The CNN is a PyTorch architecture adaptation from the locally cached QMAP implementation associated with the Witten regression approach [1, 2]. It is not a reproduction of the original author's training protocol or published performance. The fixed configuration has 183,069 parameters, dropout 0.5, Glorot initialization and 100 full-batch weighted updates per fold. Training labels are centered using the training fold only. The Keras-style Adam adaptation uses learning rate 0.001, coefficients 0.9 and 0.999, and epsilon 10^{-7}. Each fold resets seed 1729. No tuning, early stopping or validation-selected endpoint was used. There were 1,700 CNN updates in total.

The mass-unit ridge advantage reverses under peptide and record weighting. Its peptide-weighted MAE is 0.630031 versus 0.605203 for the mean, and record-weighted MAE is 0.602473 versus 0.574316. The CNN is worse under the primary MAE in both units; this statement does not claim it is worse under every reported weighting or that the architecture cannot learn on adequate data.

Right-censoring working likelihood

A later 136-ID cache supported 79 mass-unit points and 14 right bounds, and 68 molar-unit points and 12 right bounds. It admitted 173 of 232 available *E. coli* records, excluding unresolved ranges, plus/minus formats and other incompatible records. ID 12605 was withheld because its bound explicitly referred to two isolates with an unresolved observation identity. A separate retained point with ambiguous strain wording was not assumed to share identical assay conditions with another record. There were no natural left-censored observations.

The four arms were a point-only mean, point-only ridge, endpoint ridge and a Gaussian tail-likelihood ridge. They used the same composition features, training-only scaling and coefficient penalty. Endpoint and tail arms have identical rows and weights; for a right bound b and mean prediction μ, their losses are respectively $\frac{1}{2}(\mu - b)^2$ and $-\log \Phi(\mu - b)$. The Gaussian working standard deviation is fixed at one log10 unit; it is not an estimated biological noise scale. Only the point-valued observations are evaluated. Point-only controls renormalize their subset's weights and share the full training-feature scaler in this experiment, so they are not exact reruns of the earlier baseline.

There were 21 mass-unit and 13 molar-unit validation groups. Three additional molar groups contain only censored records and supply training information but no point-valued validation fold. The serial L-BFGS-B fits used the same 22-parameter initialization and ceilings of 500 iterations, 10,000 function evaluations and 40 line-search steps. All 102 fits met the fixed convergence requirement, including maximum absolute gradient at most 10^{-5}, and produced 588 point predictions.

Tail likelihood was worse than endpoint encoding under all three reported weighting schemes for

both MAE and RMSE. This negative finding concerns a misspecifiable working model and a selected point-observation subset; it does not refute censoring likelihood theory or estimate risk on latent censored outcomes. Threshold reporting may be informative, and assay covariates are incomplete. Between-group variance diagnostics cannot establish the cause of poor prediction.

Table S3e (native points). All six independently recomputed error statistics. G, P and R denote source-group/peptide, peptide-only and record weighting; units are log10 native concentration.

Unit	Model	Rows	MAE G	MAE P	MAE R	RMSE G	RMSE P	RMSE R
μg/mL	Mean	46	0.8110	0.6052	0.5743	1.0086	0.7629	0.7156
μg/mL	Ridge	46	0.7754	0.6300	0.6025	0.9475	0.7849	0.7598
μg/mL	CNN	46	0.8198	0.6272	0.5730	1.0109	0.7823	0.7204
μM	Mean	39	0.5170	0.4127	0.4409	0.6128	0.5203	0.5125
μM	Ridge	39	0.5822	0.4436	0.4529	0.6924	0.5730	0.5846
μM	CNN	39	0.5763	0.4636	0.5072	0.7115	0.6119	0.5920

Table S3e (censoring screen). All six independently recomputed error statistics. G, P and R denote source-group/peptide, peptide-only and record weighting; units are log10 native concentration.

Unit	Model	Rows	MAE G	MAE P	MAE R	RMSE G	RMSE P	RMSE R
μg/mL	Point mean	79	0.6609	0.5614	0.5241	0.8122	0.6909	0.6466
μg/mL	Point ridge	79	0.7877	0.6207	0.5811	0.9604	0.7643	0.7142
μg/mL	Endpoint	79	0.8210	0.6265	0.6089	1.0098	0.7807	0.7657
μg/mL	Tail	79	0.8441	0.6311	0.6256	1.0554	0.7963	0.8024
μM	Point mean	68	0.5496	0.5334	0.4661	0.6510	0.6329	0.5974
μM	Point ridge	68	0.5963	0.5300	0.4724	0.6893	0.6219	0.6054
μM	Endpoint	68	0.5895	0.4780	0.4763	0.7237	0.6109	0.5959
μM	Tail	68	0.7269	0.6384	0.6716	0.9022	0.7830	0.8084

Table S3d. Variance decomposition retained from the source diagnostic.

Unit	Between fraction	Between variance	Within variance	Total variance
μM	0.713730	0.269155	0.107955	0.377110
μg/mL	0.908309	0.548142	0.055333	0.603476

Tables S3a and S3b retain the original complete model/unit summaries and cohort counts; Table S3c contains all 12 tail-minus-endpoint MAE/RMSE contrasts across the three weighting schemes.

S4 Follow-up routes (Table S4)

Readout bridge and source replacement

In an initial readout experiment, we applied two readouts to the same upstream hidden states for 96 generation requests. The categorical readout gave 95/96 canonical sequences of length 1–48 and 91/96 of length 5–48, compared with 96/96 for both criteria under the scalar readout. Development MMD squared worsened in five of six source–terminal comparisons and improved in one, while

dipeptide divergence improved in all six. Those 192 readouts are paired outputs of 96 requests, not 192 independent samples. The readout intervention also changes head size and boundary decoding, preventing single-component attribution. These complementary readout effects motivated the matched reconstruction-objective experiment.

For the source-replacement analysis, we used the 32 main CE diffusion outputs (A) and added 32 requests each for training-posterior draws (B), coordinate-permuted B latents (C), and the model's prior flow with a training-derived residue/length template (D). B and D use information from the training pool; D has additional reference content unavailable under terminal-only generation. C preserves each coordinate's finite marginal values but has no single donor. All use the fixed CE decoder and legal-prefix rule.

All new requests were retained. B produced one exact training copy in the amidated group. Neither the posterior substitution nor the additional-information prior provides a fair general-generation superiority claim.

Warmup, latent-conditioned autoregression and direct latent diffusion

The warmup follow-up failed its fixed protective comparisons. Posterior-mean mean-per-sequence reconstruction declined from 0.722796 to 0.657852, and the average of eight posterior draws declined from 0.661116 to 0.589099. For posterior-mean reconstruction (M), we calculate positional residue matches divided by target length for each sequence, then average equally over the 41 development sequences. S8 averages the same equal-sequence statistic over eight posterior draws (seeds 1422–1429). The historical field name `sequence_equal` denotes equal weighting of sequences, not whole-sequence equality. These statistics differ from both the micro-residue accuracy and the whole-sequence exact-match count in the main table.

The latent-conditioned autoregressive experiment trained three arms for 1,000 updates each using cached representations, without a new encoder or diffusion pass. The matched-latent eight-draw reconstruction score was 0.690802, compared with 0.091958 for the control trained with permuted latent–sequence pairing and 0.097533 for mismatched latents. The information screen passed, while the six generation-distance comparisons failed. A no-latent autoregressive comparator C scored 0.120436 using one plain reconstruction decode with the same equal-sequence residue metric, not eight posterior draws. C has 62,501 parameters and hidden width 64, compared with 66,597 parameters in A and B; its capacity-unmatched contrast is descriptive only. This experiment made 96 new generation requests; it is distinct from the main 128-request experiment.

We trained posterior-parameter and direct-latent denoisers for 1,000 updates each and generated 64 new outputs, giving 96 comparison slots when combined with reused outputs. Direct-latent diffusion produced larger development MMD and dipeptide Jensen–Shannon distances than posterior-parameter diffusion in both terminal conditions. The mechanism criterion was therefore not met, and the sequential evaluation stopped at that stage.

Conditioning-context variants

Conditioning-context variants were not supported by the multistage screen. Equal-terminal development sequence negative log-likelihoods were 49.16431 (partial), 48.58710 (pooled), 48.91772 (separate) and 48.04302 (fresh autoregressive comparator); 384 attempts produced 381 nonempty and three empty outputs. The historical autoregressive value 57.96692 used a different protocol. Full outcomes and definitions are in Table S4. None of these exploratory results establishes improved generation under the tested protocol (Table S4).

Table S4a. Scientific endpoints of seven follow-up analyses. Source diagnostics, reconstruction analyses and generation comparisons have distinct evaluation criteria.

Route	Evaluated component	Observed result
Latent substitution	Posterior source	Development MMD increases in both terminal groups relative to CE diffusion outputs.
Coordinate permutation	Latent coordinates	Development MMD increases in both groups relative to training-posterior draws.
Reference-template prior	Reference-conditioned prior	MMD effects are mixed relative to CE; this route receives additional template information.
KL warmup	Reconstruction	Posterior-mean score decreases from 0.722796 to 0.657852; eight-draw score decreases from 0.661116 to 0.589099.
Latent-conditioned AR	Information and generation	Matched score 0.690802; mismatch 0.097533; permuted-training control 0.091958. The six generation-distance criteria are not met.
Direct-latent diffusion	Diffusion target	Development MMD and dipeptide JS increase in both terminal conditions relative to posterior-parameter diffusion.
Conditioning-context variants	Conditioning structure	Variants do not meet the combined sequence-NLL and generation criteria; 384 attempts include three empty outputs.

Table S4c. Four reconstruction criteria for the KL-warmup analysis.

Criterion	Outcome
Posterior-mean length MAE no greater than baseline	Not met
Posterior-mean sequence-equal residue accuracy does not decrease	Not met
Eight-draw sequence-equal residue accuracy improves by at least 0.05	Not met
Eight-draw supported-residue macro recall does not decrease	Not met

Table S4g. Conditioning-context attempt counts and development sequence NLL. Full CV, train/development distances and all recorded guards remain in the editable CSV.

Method	Terminal	Attempts	1–48	5–48	Empty	Dev NLL
ar_cv	Amide	32	32	31	0	44.3340
ar_cv	Free	32	30	30	2	51.7520
ar_historical	Amide	32	32	32	0	53.0354
ar_historical	Free	32	32	32	0	62.8984
ngram	Amide	32	31	20	1	45.4873
ngram	Free	32	32	27	0	53.4606
partial	Amide	32	32	21	0	45.3266
partial	Free	32	32	29	0	53.0020
pooled	Amide	32	32	25	0	44.1151
pooled	Free	32	32	23	0	53.0591
separate	Amide	32	32	19	0	44.8967
separate	Free	32	32	27	0	52.9387

Tables S4b–S4g retain all 96 source-substitution distances, ten paired warmup statistics, eight information-gate entries, both direct-latent training rows and all 12 model and terminal combinations for the conditioning-context analysis.

S5 Peptide characterization and antibacterial measurements (Table S5)

Candidate selection and assays

We tested 27 peptide candidates selected from a 1,000-candidate screen incorporating positive charge at pH 7.4, AMP-prediction score, instability index and similarity to training sequences. The selection criteria were positive net charge computed with RDKit, an external AMP-prediction score above 0.9, an instability index between 10 and 40, and sequence similarity no greater than 70% to the training set. Table S5a lists every assayed sequence and its HPLC purity. Twenty-six peptides had purities of 95.0–99.1%; pp1-7 had a purity of 80.3%.

We evaluated *E. coli* ATCC 25922, *P. aeruginosa* ATCC 27853, *S. aureus* ATCC 29213 and *E. faecalis* ATCC 29212. The 27 peptides, four strains, three replicate dilution series and ten concentrations comprise 3,240 peptide OD_{600} measurements. Blank and negative-control measurements are provided separately with the corresponding readouts. The assay panel is analyzed as a separate experiment from the reconstruction-objective comparison. The available assay records do not specify the culture medium, inoculum, incubation temperature or duration, actual solvent or the original MIC decision rule; the reported and retrospective endpoints are therefore retained separately.

Original and retrospective endpoints

Six of the 27 tested peptides had *S. aureus* MICs at or below 31.25 μg/mL in both endpoint summaries. At the strict threshold of < 200 μg/mL, seven peptides showed activity against at least one measured strain in both analyses. The *S. aureus* count at this threshold was six in the original summary and seven under the retrospective rule.

At strict *S. aureus* thresholds below 32, 64, 128 and 200 μg/mL, the original counts are six, six, six and six; the retrospective counts are six, six, seven and seven. At < 200 μg/mL, the four strain-specific counts in the original summary are 5, 1, 6 and 6; the retrospective counts are 5, 2, 7 and 5, in the strain order above. Both representations are retained in Table S5b.

Among the 78 original N cells, 73 have no numeric replicate endpoint, three have one, one has two differing numeric endpoints, and one (pp3-14 against *P. aeruginosa*) has three concordant numeric endpoints. The no-consensus cell is pp3-3 against *S. aureus*, with N/250/62.5. Table S5c reports the three differing original and retrospective endpoints; Table S5g provides complete cell-level meanings and replicate calls.

Representative active peptides include pp1-5 (`LKFTFIKGKFGKSSYY`, *S. aureus* MIC 7.8125 μg/mL) and pp2-8 (`FMPVVLTKLFGVLHFRKIY`, 15.625 μg/mL). Other peptides share these MIC values (Table S5b). Antibacterial measurements do not assess haemolysis or cytotoxicity.

Table S5a. Sequences and HPLC purity of all 27 assayed peptides.

ID	Sequence	Purity (%)
pp1-1	LNLPKILFCSAPRTRLPGNWI	95.3
pp1-2	KAFFLLYFAVPFGPWS	97.3
pp1-3	CVVSLFKSKKGGLRKHGVPFKL	95.6
pp1-4	LLWIMLAYRLAFHRNWNAGGWKLFAVV	95.1

ID	Sequence	Purity (%)
pp1-5	LKFTFIKGKFGKSSYY	96.1
pp1-7	AVCLILGLFNWGK	80.3
pp1-8	CGHLFPYWLYLGPKTPW	97.5
pp2-1	CPRTRVGSKT	95.6
pp2-2	LKLLWLIKCSG	99.1
pp2-3	AKLSFGKPFANKSKYW	97.4
pp2-4	CNNLKLHICVKKFARYYLYGPTPYY	97.5
pp2-5	RETGRGKRDKGNIGTALGCFRMI	95.2
pp2-6	VGPGKKFLPK	98.6
pp2-7	FTGGKGRDWGGHSSRKKKSLGFA	97.0
pp2-8	FMPVVLTKLFGVLHFRKIY	97.8
pp3-1	CHLKLAILFNFLKFLKQFLGNLTYV	97.2
pp3-2	HGKFFKYFNLQVCLGLARALKDGG	96.0
pp3-3	RPKQIYFFAHLRLIFGLKI	95.8
pp3-4	CLGPKLYNNWQGLGGL	97.0
pp3-5	NWQPRHFLADTFCPKLKGQILR	98.4
pp3-6	MPKGIFFALKLLWKNFEKAI	96.3
pp3-7	PKPFLWAKSDLFVKL	95.0
pp3-9	GFFPGFIVLGCKTGFTPTY	96.2
pp3-11	FKGVAKARHRLWFL	95.1
pp3-12	GFPGDVFKAKAAKA	97.7
pp3-13	KTFKKKLFLKLFYILHTPTY	95.4
pp3-14	CFANFQSKLHREARKLMGGAKKWPY	95.9

Table S5b. All 108 MIC cells in μg/mL. Each cell gives the original report followed by the retrospective algorithm. N is a nonnumeric endpoint; no-endpoint and no-consensus calls are distinguished in Table S5g.

ID	E. coli	P. aeruginosa	S. aureus	E. faecalis
pp1-1	500 / 500	N / N	250 / 125	125 / N
pp1-2	N / N	N / N	N / N	N / N
pp1-3	62.5 / 62.5	N / N	31.25 / 31.25	125 / 125
pp1-4	N / N	N / N	N / N	N / N
pp1-5	31.25 / 31.25	125 / 125	7.8125 / 7.8125	125 / 125
pp1-7	N / N	N / N	N / N	N / N
pp1-8	N / N	N / N	N / N	N / N
pp2-1	N / N	N / N	N / N	N / N
pp2-2	N / N	N / N	N / N	N / N
pp2-3	250 / 250	N / N	250 / 250	1000 / 1000
pp2-4	N / N	N / N	N / N	N / N
pp2-5	1000 / 1000	N / N	250 / 250	N / N
pp2-6	N / N	N / N	N / N	N / N
pp2-7	1000 / 1000	N / N	1000 / 1000	N / N
pp2-8	N / N	N / N	15.625 / 15.625	62.5 / 62.5
pp3-1	N / N	N / N	N / N	N / N
pp3-2	N / N	N / N	N / N	N / N
pp3-3	N / N	N / N	N / N	N / N
pp3-4	N / N	N / N	N / N	N / N
pp3-5	250 / 250	N / N	250 / 250	N / N

ID	E. coli	P. aeruginosa	S. aureus	E. faecalis
pp3-6	N / N	N / N	N / N	N / N
pp3-7	N / N	N / N	N / N	N / N
pp3-9	N / N	N / N	N / N	N / N
pp3-11	62.5 / 62.5	1000 / 1000	15.625 / 15.625	62.5 / 62.5
pp3-12	N / N	N / N	N / N	N / N
pp3-13	31.25 / 31.25	N / N	7.8125 / 7.8125	15.625 / 15.625
pp3-14	125 / 125	N / 15.625	31.25 / 31.25	N / N

Table S5c. Original and retrospective endpoints for the three differing peptide–strain combinations. Replicate calls use the retrospective rule described in Supplementary Methods.

ID	Strain	Reported	Three calculated endpoints	Algorithm
pp1-1	S. aureus	250	250/125/125	125
pp3-14	P. aeruginosa	N	15.625/15.625/15.625	15.625
pp1-1	E. faecalis	125	N/N/N	N

Table S5d contains all 3,240 OD600 readings with sample ID, strain, replicate, concentration, negative control and blank values. Tables S5e–S5g provide peptide-characterization data, threshold analyses and cell-level N semantics. The archive retains the complete source inventory and earlier contextual calculations.

S6 AMPGen as an experimental-design example (Table S6)

AMPGen combines evolutionary information and diffusion-based generation with candidate selection and antibacterial testing [3]. Its staged experimental characterization illustrates how a generative-design workflow can proceed from sequence proposals to measured activity and a broader strain panel. We use this published workflow to contextualize complementary evaluation endpoints; the computational comparator in the present study is the locally trained LSTM.

AMPGen assessed 38 peptides against *E. coli* K88 and *S. aureus* ATCC 29213; 31 met MIC ≤ 75 μM against either initial target. Nine peptides with MIC < 5 μM against both initial targets were selected for a four-strain assessment. Table S6 distinguishes these stages and the role of each experimental endpoint.

Table S6. Stages of the published AMPGen workflow and their relevance to AMP evaluation. AMPGen is a supporting experimental-design example, not a reproduced comparator in this study.

Workflow stage	Published design	Evaluation role
Generation and selection	Evolutionary information and diffusion-based generation followed by candidate selection.	Connect sequence proposals with an explicit selection stage.
Initial antibacterial assessment	38 peptides assessed against *E. coli* K88 and *S. aureus* ATCC 29213; 31 met MIC ≤ 75 μM against either target.	Measure candidate activity against a defined strain set and threshold.
Expanded characterization	Nine peptides with MIC < 5 μM against both initial targets were evaluated on a four-strain panel.	Extend initial activity screening to a broader strain profile.

The follow-up panel comprised *E. coli* ATCC 25922, *P. aeruginosa* ATCC 27853, *S. aureus* ATCC 25923 and *E. faecalis* ATCC 29212. The initial and expanded assessments are distinct stages.

Reproducibility instructions

We used the AMP model implementation at commit `e7486fb`. Experimental wrappers and configurations are provided in the AMP experiment repository at commit `f1b748d`. The accompanying repository preserves exact source versions, configuration files and dataset hashes. Analyses use the training and development partitions; the reserved evaluation partition was not accessed.

From the repository root, `python3 verify_all.py` checks packaged hashes and recalculates computational and assay summaries from the supplied outputs. It covers 960 reconstruction records, 128 main-comparison generated sequences, 96 distribution distances, 24 aligned comparisons, 255 native-unit activity predictions, 588 censoring-screen predictions, 3,240 peptide OD_{600} readings and 108 original MIC cells. This command does not retrain models or repeat inference. The supplementary command `python3 supplement/verify_supplement.py` checks table source bindings and row-level arithmetic. Architecture definitions and their source references are indexed in `source/architecture_facts.json`.

Use of generative AI tools

The authors wrote the manuscript. OpenAI Codex assisted with source and reference checking, code inspection and debugging, saved-output recalculation, factual corrections, language editing and deterministic plotting code. Claude assisted with historical coding and computational checks. The reference ledger, `verify_all.py`, `supplement/verify_supplement.py` and `manuscript/figure_source/` document the checks and plotting code. The authors checked the technical claims against these outputs and original records and retain responsibility for the scientific content. No generative image synthesis was used.

Data-file index

Tables S1–S4 contain reconstruction, generation, activity-regression and follow-up analyses. S5a displays all 27 assayed peptides; S5b–S5g contain paired MIC estimates, replicate calls, raw readings, characterization, threshold analyses and N semantics. S6 summarizes the published AMPGen workflow. Full-precision CSVs, retained source inventories and provenance hashes accompany these displays in the versioned repository.

References

[1] Lavertu A, Corbeil J, Germain P. QMAP: a benchmark for standardized evaluation of antimicrobial peptide MIC and hemolytic activity regression. *Sci Rep* 2026;16:25387. doi:10.1038/s41598-026-56004-8.

[2] Witten J, Witten Z. Deep learning regression model for antimicrobial peptide design. *bioRxiv* 2019;692681. doi:10.1101/692681. Preprint.

[3] Jin S, Zeng Z, Xiong X, et al. AMPGen: an evolutionary information-reserved and diffusion-driven generative model for de novo design of antimicrobial peptides. *Commun Biol* 2025;8:839. doi:10.1038/s42003-025-08282-7. PMID: 40447738. PMCID: PMC12125351.